

VALAREZA AREZEHGAR

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Experience

Software Developer (Part-Time)

October 2023 - May 2024

Environment and Climate Change Canada

Toronto, ON, Canada

- Revised a full-stack forecasting application by establishing communication pipelines and iteratively integrating user acceptance feedback to improve UI/UX.
- Spearheaded a complex bug fix in a configurable, regex-based Java data processor by redesigning the component's data-handling algorithm to improve modularity and reduce manual configuration steps by 50%.
- Eliminated an estimated 1-2 weeks of rework and release delay by catching 15+ production issues in DevOps pipelines and regression tests with Jenkins build artifacts and Graylog/JVM monitoring.

Software Developer Intern

May 2022 - August 2023

Environment and Climate Change Canada

Toronto, ON, Canada

- Engineered a Java component that converts daily raw ice data into structured XML/JSON by automating data ingestion, validation, and transformation, reviving a legacy forecasting system to restore 100% daily uptime.
- Designed and developed a responsive, real-time translated (EN/FR) Angular UI (NgRx/RxJS) for forecasters to publish ice forecasts used daily by mariners navigating Canadian waters to avoid hazardous ice conditions.
- Authored comprehensive component documentation, including detailed API specifications, developer READMEs, and UML diagrams to standardize knowledge transfer and accelerate onboarding.
- Leveraged Test Driven Development (JUnit 5) to lock down edge cases and prevent regressions during future changes achieving 95%+ code coverage

Projects

LUNA | C++, TypeScript, React Native, Gemini OCR, Google Cloud TTS

February 2025 - July 2025

- Assistive wearable glasses attachment that provides audio descriptions of what users are seeing.
- Built a multimodal reasoning pipeline integrating Google Gemini APIs to deliver context-aware image-to-speech narration for assistive accessibility.
- Developed C++ firmware and a React Native companion app to manage Wi-Fi and Bluetooth connectivity, enabling real-time communication between wearable and mobile systems.
- Won 1st place at QHack 2025 (\$2,440) and named a national finalist at Hult Prize Canada 2025 for innovation and social impact.
- [Link to project](#)

Undergraduate Research | C++, Python, R, Docker, Heuristic Search

May 2018 – May 2020

- Researched optimizations for AI planning in grid-based pathfinding domains, executing thousands of experiments to quantify how kinematic constraints and ray-traced collision geometry affect search efficiency.
- Designed and implemented new C++ search-guidance functions that optimized state expansion and pruned redundant paths, outperforming baseline algorithms in 90% of benchmarked tasks.
- Containerized lab planning search framework with Docker, streamlining environment setup and cutting cross-platform deployment time by 96%.
- [Link to project](#)

Education

Toronto Metropolitan University (Formerly Ryerson)

Toronto, ON, Canada

Bachelor of Science in Computer Science, Minor in Mathematics, GPA 4.23/4.33

Sep. 2020 – June 2025

- Relevant Coursework: Data Structures & Algorithms, Relational Database Systems

Technical Skills

Languages: Java, C++, TypeScript, Python, JavaScript, C#, HTML/CSS, SQL

Frameworks: Angular, React, React Native, Express.js

Developer Tools: Git, Docker, Node.js, PostgreSQL

Architectures & Methodologies: Agile, SDLC, TDD, RESTful API, Object-oriented Programming, CI/CD